

Contract: generateShortageOrder

Operation: generateShortageOrder(barcode: String, missingAmount: Integer)

Cross References: Use Case f (Issue Shortage Order)

Pre-conditions:

- The system is running and the inventory module has detected a product dropping below its minimum threshold.
- At least one supplier agreement exists in the system that provides the missing product.

Post-conditions:

- A new **Order** instance *was created* assigned to the supplier offering the lowest total price for the requested missingAmount *(instance creation)* .
- An **OrderItem** instance *was created* representing the barcode, missing amount, and calculated discount price *(instance creation)* .
- The **OrderItem** *was associated* with the newly created **Order** *(association formed)* .
- The new **Order** instance *was saved* persistently to the database *(state modification)* .